BIOT 6900 . Module 3 Assignment

Target Triage with Protein Language Models - Pancreatic Ductal Adenocarcinoma (PDAC)

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1. Re-ranking

The final ranking generally agreed with the Module 2 ranking, as most genes changed by only 0 to 2 positions, indicating that incorporating tractability evidence caused a moderate reshuffling rather than a complete change in priorities. CEL showed the largest upward movement, rising from rank 11 to rank 6, a change of +5 positions. This movement was likely influenced by its high small-molecule tractability score (p_sm = 0.72), which strengthened its overall ranking despite its middling Module 2 position. I used a weight of w = 0.6, assigning 60% importance to Module 2 biological evidence and 40% to tractability. This weighting prioritizes biological relevance while still considering whether a target is feasible for drug development. Although a different weight could change the final rankings, this approach balances the strength of the biological evidence with the practical potential for therapeutic targeting.

2. Modality

I would choose ERBB4 because it shows a sharp difference between the two modalities, with p_sm = 0.70 and p_ab = 0.08. ERBB4 encodes a receptor tyrosine kinase in the ErbB family that is located on the cell membrane. When activated by its ligand, it triggers intracellular signaling pathways involved in cell growth, survival, and differentiation. Although part of the protein is exposed outside the cell and could potentially be targeted by an antibody, its intracellular kinase domain contains an ATP-binding pocket that small molecules can bind to and inhibit. This may explain why the model favors small molecules over antibodies for ERBB4.

3. Validation

	AUROC	Average precision
Random split	0.845	0.726
Family split	0.737	0.541

Chance level for average precision = 0.334.

The family-split result is more reliable because it tests whether the model can generalize to protein families it has not seen during training. In a random split, closely related proteins may appear in both the training and test sets, allowing the model to recognize familiar sequence patterns or structural features rather than learn generalizable relationships between proteins and druggability. This data leakage can inflate the apparent performance. The family split reduces this bias by keeping related proteins together in either training or testing. The decrease in AUROC from 0.845 to 0.737 and average precision from 0.726 to 0.541 suggests that the model performs less effectively on unfamiliar protein families. Although the family-split average precision remains above the chance level of 0.334, the results

indicate that generalization to new protein families is more challenging than the random-split results suggest.

4. Variants

Gene	Variant	Note	Score	Percentile
TP53	R175H	hotspot	-5.97	19.3
APOE	C130R	rs429358, defines e4 (C112R mature)	7.21	100.0
APOE	R176C	rs7412, defines e2 (R158C mature)	-8.56	11.8
TREM2	R47H	rs75932628, AD risk variant	-5.16	47.7

The variant analysis demonstrates that evolutionary surprise does not necessarily correspond to disease-causing potential. TP53 R175H had a score of -5.97 and a 19.3rd-percentile rank, suggesting that the substitution is relatively unusual to the model. APOE R176C (e2) had a score of -8.56 and ranked at the 11.8th percentile, making it one of the most evolutionarily surprising variants in this set. In contrast, APOE C130R (e4) had a positive score of +7.21 and ranked at the 100th percentile, indicating that the model considers the mutant amino acid highly evolutionarily plausible. However, APOE e4 is a well-established Alzheimer's disease risk variant, demonstrating that evolutionary plausibility does not necessarily indicate clinical safety. Similarly, TREM2 R47H had a score of -5.16 and ranked at the 47.7th percentile, placing it near the middle of the distribution despite its association with increased Alzheimer's disease risk. These examples highlight a key limitation of sequence-based models: they assess evolutionary patterns rather than directly measuring disease risk or functional consequences. They may not capture changes in lipid transport, amyloid-beta clearance, protein interactions, or downstream biological pathways. Therefore, variant scores should be interpreted alongside clinical evidence, functional experiments, and other genetic data before drawing conclusions about pathogenicity.

5. Limitations and Next Step

Sequence-based models can provide useful information about protein function, evolutionary conservation, and druggability, but they cannot fully determine a protein's three-dimensional structure, binding-pocket geometry, accessibility, or flexibility. Therefore, a high sequence-based tractability score does not necessarily mean that a protein has a suitable binding pocket or will be an effective drug target. For Week 4, I would select MUC5B and ERBB4 for structure-based pocket assessment because they represent two different cases requiring structural investigation. MUC5B ranks first with a final score of 0.880, a small-molecule tractability score of 0.045, and an antibody tractability score of 0.545. As a large, secreted mucin glycoprotein without a typical enzyme-like binding pocket, MUC5B would benefit from structural analysis to identify accessible regions that could potentially be targeted by antibodies. ERBB4 ranks second with a final score of 0.853, a small-molecule tractability score of 0.702, and an antibody tractability score of 0.083. Because ERBB4 is a receptor tyrosine kinase with an intracellular ATP-binding pocket, it is a plausible small-molecule target, but structural assessment is still needed to evaluate the pocket's geometry and suitability for drug binding. Comparing these proteins would help determine whether their sequence-based predictions translate into realistic structural opportunities for drug development. Finally, zero genes were unmapped in Step 1, so no additional evidence is needed to resolve unmapped targets.

Final Ranked Targets (Top 6 of 15)

Rank	Gene	week2_score	p_sm	p_ab	final
1	MUC5B	0.976	0.045	0.545	0.880
2	ERBB4	0.973	0.702	0.083	0.853
3	TNS4	0.975	0.214	0.003	0.733
4	COL11A1	0.984	0.096	0.000	0.707
5	FN1	0.971	0.477	0.292	0.653
6	CEL	0.954	0.717	0.009	0.600